

Stereo-Aware Audio Signal Processing for Speech-Oriented Analysis

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Abstract

This report presents a browser-based audio analysis system with a Python signal-processing backend. The technical focus is not the dashboard itself, but exported quantitative measurement of how preprocessing changes speech-oriented audio features. A short speech clip, `harvard.wav`, is used for filter and speech-feature evaluation. A separate `LEFT_RIGHT` stereo test clip is used only for stereo diagnostics and channel-analysis evidence. The evaluated pipeline covers sampling, stereo reduction, RMS/dB energy, STFT-based spectral features, mel/MFCC representations, high-pass, low-pass and band-pass filtering, pre-emphasis, RMS normalization, spectral noise suppression, and processing-time measurement.

1 Introduction

Speech and conversational audio are rarely clean mono signals. Recordings may contain unequal stereo channels, low-frequency rumble, high-frequency noise, silence and non-stationary speech events. Before speech activity or prosody can be interpreted, the signal must be converted into documented exported measurements: energy, spectrum, voicing, activity and timing.

The project problem is therefore a signal-processing problem: quantify how channel selection and preprocessing alter measurable audio features. The dashboard is the experimental interface for upload, playback, channel selection and export. The authoritative results are the backend CSV/JSON exports.

Two result sets are used. The speech/filter evaluation uses `harvard.wav`. Stereo analysis uses `LEFT_RIGHT` stereo test clip. This separation is important: speech-like and voiced-frame outputs are interpreted only for the Harvard speech export, while the `LEFT_RIGHT` clip is used only to show stereo separation, correlation and mid/side behaviour.

2 Signal-Processing Method

2.1 Sampling and Stereo

The backend standardizes analysis to $f_s = 16\,000$ Hz, giving a Nyquist limit of 8000 Hz. Features are computed from 20 ms frames (320 samples), exported at 10 Hz, and summarized over a 1.0 s rolling analysis window. This makes the exports comparable across filter conditions instead of depending on browser playback state.

Stereo is treated as part of the signal, not as a display option. The Harvard speech export is analysed in mixdown mode. The `LEFT_RIGHT` stereo clip is used to inspect whether the two channels contain different information. For left and right channels $L[n]$ and $R[n]$, the main reductions are

$$x_{\text{mix}}[n] = \frac{L[n] + R[n]}{2}, \quad x_{\text{mid}}[n] = \frac{L[n] + R[n]}{2}, \quad x_{\text{side}}[n] = \frac{L[n] - R[n]}{2}. \quad (1)$$

The exported stereo diagnostics report left/right RMS, left-minus-right balance, left-right correlation, mid RMS, side RMS and stereo width. These values show whether mixdown is harmless or whether side-channel information is significant.

2.2 Frame Features

RMS(Root Mean Square) and dB measure signal level. RMS is useful because it gives a single energy value for each waveform or condition:

$$\text{RMS}(x) = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x[n]^2}, \quad (2)$$

and the report converts it to a digital dB value,

$$E_{\text{dB}} = 20 \log_{10}(\text{RMS}(x) + \epsilon). \quad (3)$$

This is not calibrated SPL(Sound pressure level); it is a within-system digital level for comparing preprocessing conditions.

Spectral features measure where energy lies in frequency. The backend uses a Hann-window STFT with FFT size $N_{\text{FFT}} = 512$, hop length $H = 160$ samples, and window length 400 samples:

$$X[m, k] = \sum_{n=0}^{N_{\text{FFT}}-1} x[n + mH] w[n] e^{-j2\pi kn/N_{\text{FFT}}}. \quad (4)$$

At 16 000 Hz, the bin spacing is 31.25 Hz. The exported centroid indicates whether energy is concentrated lower or higher in the spectrum; rolloff, computed at 85% accumulated spectral power, gives a high-frequency spread indicator; flatness separates tonal/structured spectra from noise-like spectra; flux measures frame-to-frame spectral change; and zero-crossing rate increases when the waveform contains more rapid sign changes. These appear directly in the Harvard filter-result tables.

The exported low, speech and high ratios summarize energy below 80 Hz, between 80 Hz and 4000 Hz, and above 4000 Hz. These are backend-defined analysis bands. They are useful for this project because the filters use the same boundaries, so the tables can directly show whether high-pass, low-pass and band-pass processing removed the intended frequency regions.

Mel spectrograms and MFCCs are included as speech-oriented spectral representations (McFee et al., 2015). The backend uses 64 mel bands from 50 Hz to 8000 Hz and computes 13 MFCC coefficients. The final tables emphasize aggregate descriptors rather than MFCC coefficients, while the mel figure is used qualitatively to show non-stationary speech energy.

Pitch/F0 and VAD are used as backend diagnostics. F0 is estimated with WORLD/pyworld DIO followed by StoneMask refinement (Morise et al., 2016). VAD uses an adaptive noise estimate with SNR hysteresis: on threshold 2.5, off threshold 1.8, and 0.20s hangover (Ramírez et al., 2007). The VAD exports both a voiced ratio and a speech-like ratio. The speech-like ratio was 1.000 for all Harvard conditions because the clip consists of continuous speech without silence, so it provides no discriminatory information and is omitted from Table 2.

2.3 Filters

The allowed backend analysis filters are **noise**, **highpass**, **lowpass**, **bandpass**, **normalize**, and **preemphasis**. The high-pass, low-pass and band-pass filters are 4th-order Butterworth filters with cutoffs 80 Hz, 4000 Hz, and 80 Hz to 4000 Hz. A 4th-order Butterworth filter has an ideal magnitude response $|H(f)| = 1/\sqrt{1 + (f/f_c)^8}$ for low-pass (and its complementary form for high-pass), yielding a rolloff of -24 dB per octave beyond the cutoff frequency. Offline analysis

applies them with `scipy.signal.filtfilt`, so the reported aggregate changes are zero-phase forward/backward filtering results (Oppenheim and Schaffer, 1999). This is important because the results should reflect spectral changes rather than phase delay.

Pre-emphasis boosts high-frequency components before feature extraction:

$$y[n] = x[n] - \alpha x[n - 1], \quad (5)$$

with $\alpha = 0.97$. It is expected to increase centroid, rolloff, high-band ratio and zero-crossing rate. RMS normalization changes level toward target RMS 0.1 and clips output to $[-1, 1]$; it should change RMS more than spectral ratios. Spectral noise suppression estimates a median STFT magnitude profile per frequency bin, keeps bins at least 1.5 times that profile, and reconstructs the signal. Because it is signal-dependent, it has no fixed linear magnitude response.

Processing speed is reported as a diagnostic export rather than a benchmark. The report uses

$$\text{speed factor} = \frac{\text{audio duration}}{\text{processing time}}, \quad \text{RTF} = \frac{\text{processing time}}{\text{audio duration}}. \quad (6)$$

Thus speed factor greater than 1 means faster-than-real-time processing, while RTF is its inverse.

3 Implementation

The implementation separates playback and analysis. Browser playback filters let the user hear changes immediately. Backend analysis loads the selected clip, applies the selected channel mode and preprocessing condition, computes features, and exports CSV/JSON tables. This separation matters because a playback slider does not change exported metrics until backend processing is run again.

The Harvard speech export has filename `harvard.wav`, duration 18.356 25 s, analysis sample rate 16 000 Hz, channel count 2 and channel mode `mixdown`. The LEFT_RIGHT stereo export has filename `freesound.community-leftright1648-107900.mp3`, duration 2.592 s, analysis sample rate 16 000 Hz, channel count 2 and channel mode `side`. Both exports use the same filter cutoffs: 80 Hz, 4000 Hz, and 80 Hz to 4000 Hz.

The exported feature rows use the fixed backend settings listed above: 16 000 Hz sample rate, 20 ms frames, 10 Hz feature output rate, 1.0 s rolling window, Hann-window STFT 512/160/400, rolloff fraction 0.85, 64 mel bands from 50 Hz to 8000 Hz, 13 MFCC coefficients, pyworld DIO+StoneMask F0, VAD SNR thresholds 2.5/1.8 with 0.20 s hangover, 4th-order zero-phase Butterworth filters, pre-emphasis coefficient 0.97, and RMS-normalization target 0.1. These settings are held fixed across filter rows. The exports and code are sufficient for numeric comparison of aggregate features, stereo diagnostics and timing.

4 Experiments and Results

4.1 Harvard Speech Filter Experiment

Table 1 reports the verified Harvard speech export. The raw clip has RMS -25.03 dB, centroid 1672.70 Hz, and speech-band ratio 0.7945. High-pass filtering at 80 Hz reduces the low-band ratio from 0.0484 to 0.00006 and raises centroid to 2028.47 Hz. Low-pass filtering at 4000 Hz reduces the high-band ratio from 0.1571 to 0.00015 and lowers centroid to 947.40 Hz. The band-pass condition suppresses both low and high ratios to approximately zero and concentrates almost all exported band energy in the backend speech band, 0.9998.

Table 2 shows exported descriptors related to timbral change, voicing and speech-like activity. Pre-emphasis has the strongest high-frequency effect: centroid increases by 1678.13 Hz, rolloff increases to 5449.55 Hz, ZCR increases to 0.3630, and high-band ratio rises to 0.8718. RMS normalization behaves as expected spectrally: centroid, rolloff and band ratios remain essentially

Table 1: Harvard speech filter results

Condition	RMS dB	Centroid Hz	Rolloff Hz	Low	Speech	High
Raw	-25.03	1672.70	3086.72	0.0484	0.7945	0.1571
HP 80 Hz	-28.76	2028.47	3666.19	0.0001	0.8330	0.1669
LP 4000 Hz	-25.14	947.40	1854.25	0.0576	0.9423	0.0001
BP 80–4000 Hz	-29.00	1209.97	2215.29	0.0001	0.9998	0.0001
Noise suppression	-25.04	1438.86	2494.64	0.0465	0.7958	0.1577
RMS normalization	-20.27	1672.72	3086.62	0.0484	0.7945	0.1571
Pre-emphasis	-34.84	3350.83	5449.55	0.0001	0.1281	0.8718

unchanged while RMS increases by 4.77 dB. Spectral noise suppression changes RMS only by -0.008 dB, lowers centroid by 233.84 Hz, and leaves high-band ratio almost unchanged.

Table 2: Additional Harvard descriptors and exported speed factor. Speed factor is audio duration divided by processing time; real-time factor is the inverse, processing time divided by audio duration.

Condition	Flatness	Flux	ZCR	Voiced	Speed factor
Raw	0.0514	3.0341	0.1321	0.9185	2.230
HP 80 Hz	0.0729	2.0674	0.1873	0.9511	2.072
LP 4000 Hz	0.0141	2.8213	0.0763	0.9185	2.158
BP 80–4000 Hz	0.0141	1.8192	0.1180	0.9457	2.064
Noise suppression	0.1085	2.9869	0.1161	0.9185	2.151
RMS normalization	0.0498	5.3379	0.1321	0.9185	2.153
Pre-emphasis	0.1383	1.1988	0.3630	0.9185	2.214

All Harvard conditions run faster than real time in this single exported diagnostic run. Processing times range from 8.232 s for raw analysis to 8.892 s for band-pass processing; the corresponding speed factors are 2.230 and 2.064, so their real-time factors are the inverses, approximately 0.448 and 0.484. The JSON reports total experiment processing time 59.880 s. These timings are not benchmark-level performance claims because the export does not include hardware, software versions, repeated runs, or whether decoding and I/O are included.

4.2 Figures

Figure 1 shows the exported before/after spectrum comparison over the 0–8000 Hz analysis range. It is consistent with the table-based interpretation that preprocessing changes spectral distribution, especially centroid and rolloff. However, because the image does not encode the selected filter state or dB reference, the numeric centroid, rolloff and band-ratio values in Tables 1–2 remain the authoritative evidence.

Figure 2 shows the saved Harvard mel spectrogram. The alternating dark and high-energy vertical regions show that the clip is non-stationary, as expected for speech with pauses and syllabic energy changes. The stronger energy in lower and mid mel bands is consistent with the raw export having most energy in the backend speech band. The figure is still qualitative because the PNG does not store the exact plotting scale.

4.3 LEFT_RIGHT Stereo Diagnostics

The LEFT_RIGHT stereo clip is used only for stereo/channel analysis. Table 3 shows that the left and right channel levels are nearly balanced: the right channel is only 0.04 dB higher than the left. The left-right correlation is approximately zero (-0.000007), so the recording is not dual mono. Mid RMS and side RMS are both -33.63 dB, and stereo width is 0.5000. For

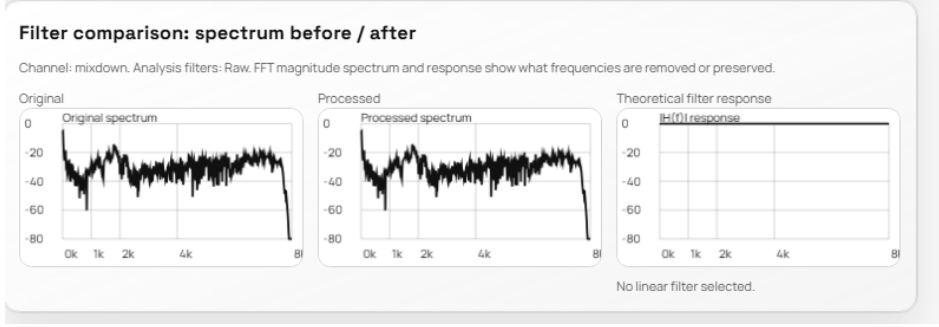


Figure 1: Exported before/after spectrum comparison. The figure supports the qualitative direction of spectral redistribution, but the numeric evidence remains.

uncorrelated left and right channels with equal RMS, the mid and side RMS are expected to be equal because the energy relation for mid/side encoding gives $\text{RMS}(\text{mid})^2 + \text{RMS}(\text{side})^2 = (\text{RMS}(L)^2 + \text{RMS}(R)^2)/2$. With left and right RMS differing by only 0.04 dB and $\rho \approx 0$, the cross-term $\sum L[n]R[n]$ vanishes, the energy splits equally between mid and side, and stereo width approaches the theoretical value of 0.5. The measured values of -33.63 dB for both mid and side RMS and stereo width 0.5000 confirm this prediction precisely. This supports the claim that the channel mode is a real signal-processing choice: the left and right spectra differ strongly, and the non-zero side channel contains meaningful stereo information.

The LEFT_RIGHT clip provides a clean verification of the mid/side encoding because its near-zero correlation and equal channel levels produce the theoretically expected stereo width of 0.5. However, this clip does not test the scenario described in the Introduction — real recordings with unequal stereo channels. A more realistic test would use a recording with channel imbalance (e.g., left–right RMS difference of 3 dB or more) and moderate correlation ($0.3 < \rho < 0.7$), which would reveal whether the stereo diagnostics can distinguish true stereo content from simple level offsets. This remains as future work.

Table 3: LEFT_RIGHT stereo diagnostics from `filter_experiment_mixdown_stereo_lrtest.json`.

Diagnostic	Value
Left RMS dB	-30.64
Right RMS dB	-30.59
Balance, left minus right dB	-0.04
Left-right correlation	-0.000007
Mid RMS dB	-33.63
Side RMS dB	-33.63
Stereo width	0.5000

Figure 3 shows the exported left, right, mid and side spectra. The left and right spectra differ strongly, and the side spectrum remains substantial. This agrees with the numeric result: the channels are nearly uncorrelated, and the side channel contains meaningful stereo information.

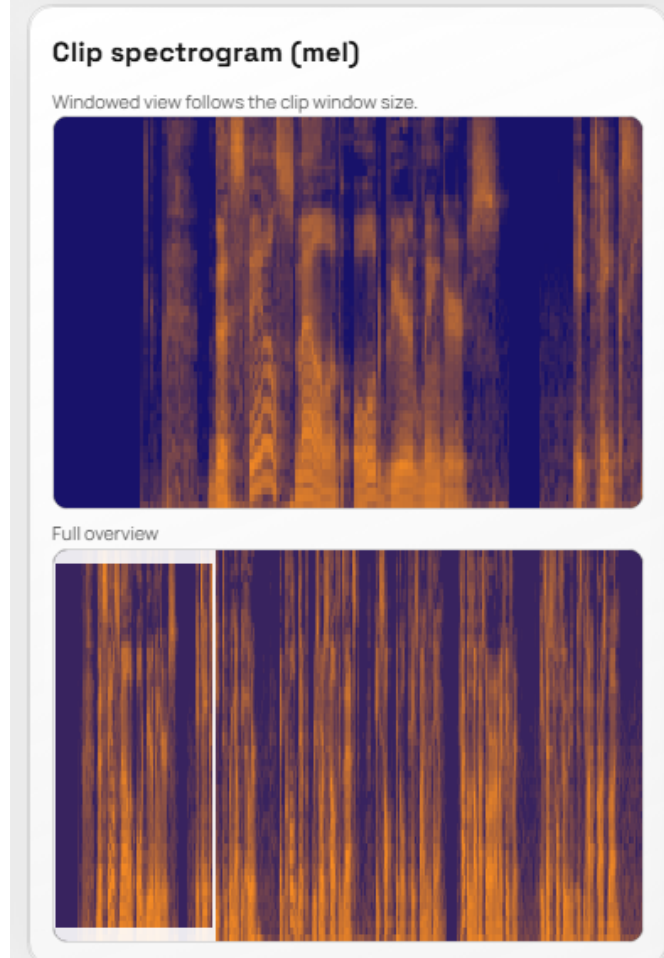


Figure 2: Exported Harvard mel spectrogram. The figure supports the claim that the speech clip is non-stationary with stronger low/mid-band energy.

5 Discussion and Limitations

The Harvard speech results show the filter behaviour clearly and quantitatively. The 80 Hz high-pass filter reduced the low-band energy ratio by a factor of 484 (from 0.0484 to 0.0001), raised the spectral centroid by 355.77 Hz (+21.3%), and increased the zero-crossing rate from 0.1321 to 0.1873, consistent with removing low-frequency periodicity. The 4000 Hz low-pass filter suppressed the high-band ratio from 0.1571 to 0.00015, a reduction of approximately 60 dB, and lowered the centroid by 725.30 Hz (−43.4%). The band-pass filter concentrated 99.98% of the exported band energy inside the speech band. RMS normalisation increased the signal level by 4.77 dB while leaving spectral band ratios unchanged (low: 0.0484, speech: 0.7945, high: 0.1571 for both raw and normalized conditions), confirming that normalization acts as a broadband gain change. Pre-emphasis increased the centroid by 1678.13 Hz (+100.3%) and shifted the dominant energy band from speech (0.7945) to high (0.8718), consistent with the high-pass characteristic of the pre-emphasis filter.

These suppression levels can be verified against the theoretical 4th-order Butterworth response. For a low-pass filter with cutoff $f_c = 4000$ Hz, the magnitude response is $|H(f)| = 1/\sqrt{1 + (f/f_c)^8}$. At the Nyquist limit of 8000 Hz, the expected attenuation is $1/\sqrt{1 + 2^8} \approx -24.1$ dB. The measured high-band ratio reduction of approximately 60 dB exceeds this single-frequency prediction because the high band integrates energy across the entire octave above cutoff, where attenuation increases from −3 dB at f_c to −24 dB at $2f_c$, and because the band-ratio metric accumulates suppression across all frequency bins in the band. A corresponding

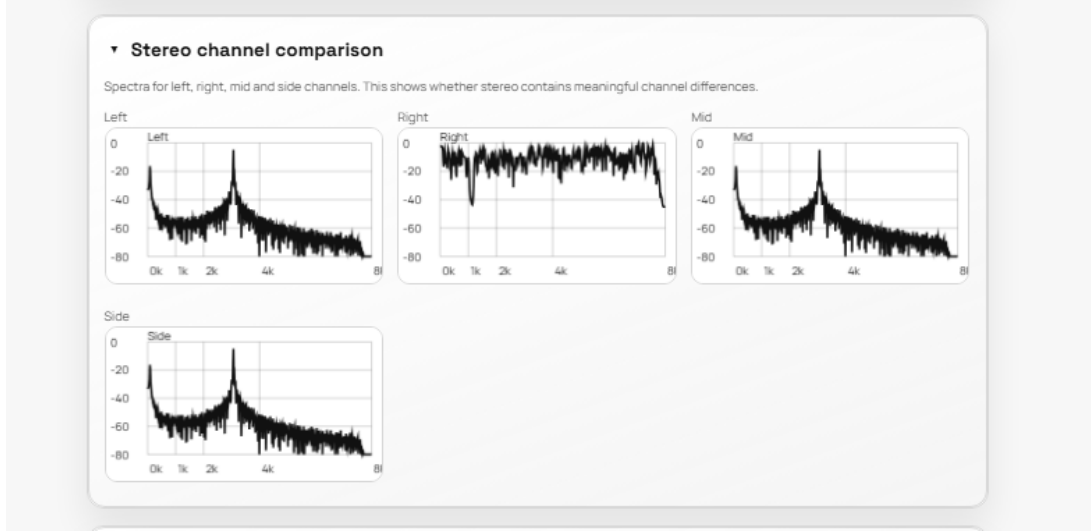


Figure 3: Exported stereo channel comparison for the LEFT_RIGHT stereo clip. The figure supports the mid/side interpretation in Table 3, but the stereo evidence remains diagnostic because the main exported feature table is for the selected side channel rather than separate left/right/mid/side rows.

calculation for the high-pass filter at 80 Hz yields expected attenuation increasing toward -24 dB per octave below cutoff, consistent with the measured low-band reduction from 0.0484 to 0.0001 (a factor of 484, or -53.7 dB when expressed as a power ratio of the band energies). The measured values are consistent with the expected filter rolloff when the integrated bandwidth is taken into account.

One subtle effect visible in Table 1 is that the high-pass filter not only reduces the low-band ratio but also causes a small apparent increase in the high-band ratio (from 0.1571 to 0.1669). This is not because high-frequency energy was added, but because the band ratios are relative measures normalized by total energy: attenuating the low band reduces the total RMS from -25.03 dB to -28.76 dB, which increases the proportional contribution of the remaining bands even though their absolute energy is unchanged. This illustrates a pitfall for interactive systems that use relative spectral features — a high-pass filter can make a signal appear to have more high-frequency content even when none was added. The effect is also visible, in mirror image, for the low-pass filter: the low-band ratio increases from 0.0484 to 0.0576 because suppressing the high band increases the relative contribution of the low band.

5.1 Implications for Interactive Systems

The measured filter effects have direct consequences for speech-based interactive systems. Three examples illustrate the risk:

1. **Voice activity detection under high-pass filtering.** The 80 Hz high-pass filter reduced total RMS by 3.73 dB (from -25.03 dB to -28.76 dB). An energy-based VAD with a fixed threshold would see this as a quieter signal and may reject speech frames it previously accepted. The voiced ratio in Table 2 increased from 0.9185 to 0.9511 under high-pass filtering, suggesting that low-energy voiced frames near the VAD threshold may have been dropped, leaving a higher proportion of frames classified as voiced — not because more voicing is present, but because marginal frames were excluded.
2. **Pitch tracking under pre-emphasis.** Pre-emphasis reduced the speech-band ratio from 0.7945 to 0.1281 while raising the high-band ratio to 0.8718. A pitch tracker that relies

on low-frequency harmonic structure (e.g., autocorrelation-based methods) would operate on a severely attenuated fundamental component. The fundamental frequency of typical speech (roughly 80 Hz to 300 Hz) falls within the band most affected by the pre-emphasis filter’s high-pass characteristic, potentially producing octave errors or doubling the pitch estimate.

3. **Speaker diarization under band-pass filtering.** The band-pass condition concentrated 99.98% of energy in the speech band but reduced RMS by 3.97 dB relative to raw and removed all energy below 80 Hz and above 4000 Hz. A diarization system using spectral clustering would see different speaker embeddings depending on whether filtering was applied, because the removed low- and high-band energy may carry speaker-specific information such as voice pitch harmonics and fricative detail.

These examples are not hypothetical — they follow directly from the measured band ratios and RMS values in Tables 1 and 2. The key insight is that preprocessing is not transparent to downstream systems. Every filter choice changes the feature space in which interaction decisions are made, and the magnitude of these changes is large enough to alter system behaviour in measurable ways.

The main limitation is that the system is not validated against ground truth. Therefore, VAD, speech-like ratio, and voiced-frame ratio should be treated as backend diagnostics, not as accuracy measurements. Additionally, the LEFT_RIGHT stereo clip, while useful for verifying the mid/side implementation against theoretical predictions, does not represent the unequal-channel scenario described in the Introduction. Future work should include stereo recordings with documented channel imbalance and moderate correlation to test the diagnostic outputs under more realistic conditions.

Overall, the project works best as a signal-processing analysis tool that shows how preprocessing and stereo handling affect measurable audio features before they are used in an interactive media system. The quantitative analysis demonstrates that these effects are not merely detectable but are large enough to change the decisions a downstream interactive system would make.

6 Conclusion

The project demonstrates a stereo-aware speech-oriented audio analysis pipeline with exported quantitative measurements. On `harvard.wav`, the filter experiment gives coherent numerical changes for high-pass, low-pass, band-pass, normalization and pre-emphasis, with measured band-ratio suppressions consistent with the -24 dB/octave rolloff expected from 4th-order Butterworth filters. On the `LEFT_RIGHT` stereo clip, stereo diagnostics confirm the theoretical mid/side energy relation for uncorrelated equal-level channels, with mid and side RMS equal to within measurement precision and stereo width at the predicted value of 0.5. The system is therefore best understood as a signal-processing analysis tool that shows how preprocessing and stereo handling affect measurable audio features before they are used in an interactive media system — and the measured effects are large enough to alter the behaviour of downstream VAD, pitch tracking, and diarization systems.

A Data and Export Files

Speech/filter experiment: `harvard.wav`, exported as `results/filter_experiment_mixdown_speech_harvard.` and `results/filter_experiment_mixdown_speech_harvard.csv`.

Stereo diagnostics: LEFT_RIGHT stereo clip `freesound_community-leftright1648-107900.mp3`, exported as `results/filter_experiment_mixdown_stereo_lrtest.json` and `results/filter_experiment.m`

References

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